

Predictive Maintenance for Metro Climate and Air Systems: A Cycle-Level Data Mining Pipeline for the MetroPT-3 Sensor Stream

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ABSTRACT

Unplanned failure of a metro train’s Air Production Unit (APU) removes the vehicle from service, degrades cabin climate control, and drives a large, avoidable increase in auxiliary energy consumption. This proposal sets out a complete data mining pipeline over the MetroPT-3 dataset, comprising 1,516,948 multivariate readings sampled at 0.1 Hz from seven analogue and eight digital sensors on an operational Porto Metro compressor between 1 February and 1 September 2020, against which four air-leak failures are documented in the operator’s maintenance reports.

Profiling of the raw file establishes both the opportunity and the difficulty. The compressor runs under load for 14.6% of normal operation but 99.6% of the time inside the reported failure windows, a 6.8-fold increase in duty cycle that is directly attributable to air loss. However, only 1.98% of records fall inside a labelled window, and roughly three quarters of the hours in which the 24-hour rolling duty cycle exceeds 0.8 lie outside every reported event. A naive threshold detector consequently attains recall 0.66 at precision 0.21 ($F_1 = 0.32$), and that figure is adopted here as the baseline any proposed method must beat. We therefore argue that the operative problem is not detection but discrimination: separating leak-driven load from demand-driven load. The proposed pipeline addresses this by moving the unit of analysis from the 10-second reading to the compression cycle, of which 16,923 are present on raw transitions, reducing the working set by roughly two orders of magnitude while preserving the physical quantity that encodes a leak. Three modelling tasks are specified against distinct label budgets (clustering, semi-supervised anomaly detection, lead-time classification) and evaluated with event-level precision, recall and lead time under a fixed alert budget, alongside throughput and latency measured over the full 1.5-million-row stream.

CCS CONCEPTS

- **Computing methodologies** → *Anomaly detection*; Machine learning; Cluster analysis;
- **Information systems** → *Data stream mining*;
- **Applied computing** → *Transportation*.

KEYWORDS

predictive maintenance, anomaly detection, multivariate time series, railway systems, sensor data, class imbalance, concept drift

1 INTRODUCTION

A metro train’s Air Production Unit compresses and dries the air that feeds the pneumatic circuit behind climate control, door actuation, suspension levelling and braking assistance. Degradation therefore surfaces first as a comfort complaint, cabin temperature drifting out of band or doors cycling slowly, and only later as a mechanical fault. Out-right failure withdraws the train mid-line, and recovery takes hours on a timetable with little spare capacity.

The economics follow the same asymmetry. Calendar-based servicing, the prevailing practice, is wrong in both directions at once: it replaces healthy components on a fixed schedule and it still misses faults that develop between intervals. Condition-based maintenance promises to fix this, but only if a model can convert raw sensor streams into an alert that arrives early enough to schedule around and precisely enough that a technician is not dispatched to a healthy unit.

1.1 The dataset and what profiling reveals

The MetroPT-3 dataset [14] is an unusually unforgiving test bed for that promise. It contains 1,516,948 readings taken every ten seconds from an operational Porto Metro compressor over 213 consecutive days, across seven analogue channels (pressures in bar, oil temperature in °C, motor current in A) and eight digital state flags. Four air-leak events are documented in the operator’s own maintenance reports, listed in Table 1.

We profiled the file itself, and its published description turns out to understate several things. The digital flag COMP is asserted low, so COMP = 0 marks the compressor operating under load. Across normal operation the unit is in that state 14.63% of the time; aside from the four reported windows it is in that state 99.56% of the time. That 6.8-fold in-

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Table 1: Documented failure events. All four are air leaks classified as high stress. Note that the source documentation attaches a maintenance note dated 30 April to the event of 29–30 May, which we treat as a transcription error in the label metadata and not a data error.

#	Start	End	Duration
1	2020-04-18 00:00	2020-04-18 23:59	24.0 h
2	2020-05-29 23:30	2020-05-30 06:00	6.5 h
3	2020-06-05 10:00	2020-06-07 14:30	52.5 h
4	2020-07-15 14:30	2020-07-15 19:00	4.5 h

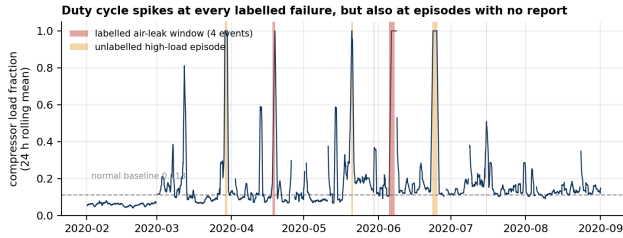


Figure 1: Twenty-four-hour rolling compressor duty cycle across the full 213-day record. Every labelled failure coincides with a duty-cycle excursion, but several excursions of comparable magnitude carry no maintenance report. This is the defining difficulty of the dataset.

crease is the physical signature of an air leak: the unit must run almost continuously to hold a reservoir pressure it can no longer maintain. Oil temperature rises 21% and motor current 179% over the same comparison, all consistent with sustained loading.

Figure 1 shows why this is not a solved problem. Every labelled failure sits on a duty-cycle spike, but so do several intervals with no report attached. Quantifying this: of the hours in which the 24-hour rolling duty cycle exceeds 0.8, 74.8% fall outside every labelled window. A threshold detector tuned on this signal reaches recall 0.66 at precision 0.21, for $F_1 = 0.32$. Whether those unlabelled excursions are unreported faults, tolerated minor leaks, or simply days of exceptional pneumatic demand is not recoverable from the data alone. The evaluation design in Section 4 therefore treats those episodes as a separate category instead of scoring them as errors.

1.2 Contribution

This proposal makes three commitments that distinguish it from prior work on the same data.

- (i) **The compression cycle, not the reading, is the unit of analysis.** A leak barely perturbs any individual reading. What it changes is how long the compressor must run to refill the reservoir and how soon it must run again. Aggregating the 1.5 M readings into the roughly 16,900 load cycles present in the record makes that quantity directly measurable and shrinks the working set by roughly two orders of magnitude.
- (ii) **Evaluation is event-level and budget-constrained.** We

report whether each of the four events was caught, with what lead time, and at what cost in false alarms per 30 days. Per-sample accuracy is uninformative at 98% negative prevalence.

- (iii) **An explicit, published baseline.** The naive duty-cycle threshold reaches $F_1 = 0.32$ at the per-sample grain, and that figure is stated up front so that no later result can be presented against a quietly chosen comparator. The threshold is recomputed at the modelling grain before any comparison is drawn, since a per-sample figure and a cycle-level one are not like for like.

2 RELATED WORK

Prior research falls into four lines, each contributing a component and leaving a gap this proposal addresses.

2.1 Physics-based and rule-based diagnostics

The oldest line of work in HVAC and compressed-air fault detection compares observed behaviour against an analytical or expert model of the healthy system, flagging residuals that exceed a tolerance. The ASHRAE RP-1043 chiller fault programme [5] is the canonical reference, and rule-based detection remains standard in deployed building management systems; data-driven fault detection in industrial compressed air systems more broadly is treated by Wenzel et al. [16]. The alerts are interpretable by construction and require no labelled failures. The cost is that thresholds are calibrated to one unit under one duty profile, which is why we do not adopt this as the primary method. Figure 1 shows the failure mode that produces on real operational data, where demand-driven load excursions are indistinguishable from fault-driven ones by magnitude alone.

2.2 Deep anomaly detection on the APU

Davari et al. [6] applied sparse autoencoders and LSTM autoencoders to this compressor, treating reconstruction error over a sliding window as the anomaly score, and Davari et al. [7] extended the framing to fault-type discrimination. These works establish that the signal is learnable and set a performance reference. Two limitations are why this study is not a replication of that one. First, reconstruction error is a scalar with no decomposition, so an operator receives a flag without an attribution to a sensor or a mechanism. Second, evaluation is conducted largely at the sample level, which on a stream with 10-second autocorrelation rewards a model for repeatedly re-detecting the same event.

2.3 Streaming and explainable predictive maintenance

Barros et al. [1] address failure detection for this APU in an operational streaming context, and Gama et al. [8] survey the drift-aware online setting that industrial deployment demands. Ribeiro et al. [12] supply the sequence-mining and rule-extraction machinery for turning stream patterns into human-readable statements. This group supplies our deployment framing and our concern with concept drift, which the monthly statistics confirm is present: mean duty cycle rises

from 0.059 in February to 0.297 in June before falling back to 0.131 in August. What this literature generally does not do is exploit the cyclic structure of a reciprocating compressor, which is where we position the contribution.

2.4 Imbalance and evaluation methodology

Because 1.98% of records are positive, the evaluation design draws on the imbalanced-learning literature. Saito and Rehmsmeier [13] establish that precision-recall curves are more informative than ROC under heavy skew, which is why the evaluation below specifies PR-AUC throughout. Bergmeir and Benítez [2] give the case for blocked cross-validation on temporally dependent data, and Lundberg and Lee [10] supply the attribution method that converts a fired alarm into a statement about which sensors drove it.

3 PROPOSED METHODOLOGY

The pipeline runs in five stages. Each subsection states the decision taken, the alternative rejected, and the reason.

3.1 Data Understanding

The purpose of this stage is to characterise the normal operating envelope before any model is fitted, so that later anomaly scores have a defensible reference.

Distributional analysis. Per-channel summaries, histograms and empirical CDFs are computed separately for each digital state, because the marginal distribution of an analogue channel is a mixture over operating modes and summarising it unconditionally is misleading. TP2, for instance, has mean 1.37 bar overall, but that figure reflects a bimodal mixture of idle and loaded operation and describes neither state.

Correlation and dimensionality. The analogue correlation matrix (Figure 2) reveals near-exact redundancy: TP3 and Reservoirs correlate at $r \approx 1.000$ to three decimals, and PCA confirms this, with the seventh component carrying negligible variance and five components accounting for 99.8%. We drop one of the pair, because collinear inputs destabilise tree-based feature attribution and would split importance arbitrarily between two names for the same measurement. TP2 and H1 correlate at -0.96 , which is physically expected and retained as an informative pair.

Temporal structure. Autocorrelation and spectral analysis of the duty cycle series characterise the daily service rhythm, which must be separated from degradation trend. Approximately 79 load cycles occur per day. Cycle segmentation is performed on COMP transitions and validated by hand against a random sample of 50 cycles before any downstream stage consumes it. The count of 16,923 quoted throughout is taken on raw COMP transitions and is therefore an upper bound: the debouncing step in Section 3.2 will merge flicker-split fragments, and the gap and frozen-channel exclusions will remove further cycles, so the working count entering the modelling stage will be lower. The reduction figure moves with it and is quoted on the raw count.

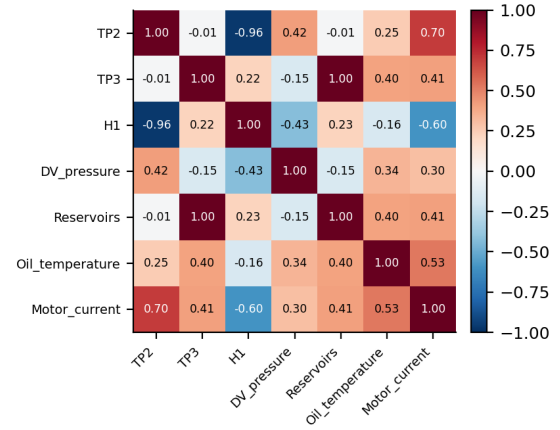


Figure 2: Pearson correlation among the seven analogue channels. TP3 and Reservoirs are the same measurement under two names; the strong negative coupling of TP2 and H1 reflects the cyclonic separator discharge.

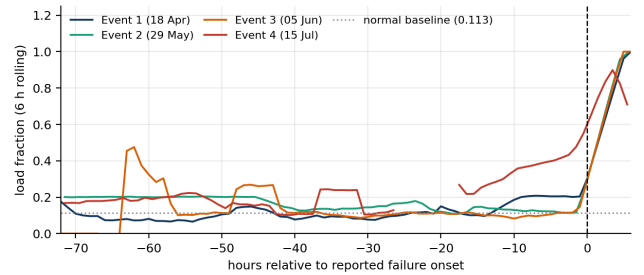


Figure 3: Six-hour rolling duty cycle in the 72 hours before each documented failure. Escalation precedes every event, but its onset ranges from more than two days to less than six hours.

Outlier and precursor analysis. Figure 3 plots the six-hour rolling duty cycle in the 72 hours preceding each event. Mean load fraction in the final six hours before onset is 0.29, 0.22, 0.27 and 0.55 against a normal baseline of 0.11, so escalation is present in all four cases, but a fixed two-times-baseline trigger fires 62 hours ahead for event 3, 36.5 hours ahead for event 4, and not at all within 72 hours for events 1 and 2. Lead time is therefore highly variable, and reporting a single mean lead time without its spread would misrepresent the result.

3.2 Pre-processing and Management

Sensor data of this kind is fuzzy in three distinct ways. No single blanket cleaning step addresses all three, so each is handled separately.

Timestamp irregularity. Nominal sampling is every ten seconds, but 128,277 intervals are nine seconds, 38,321 are twelve, and 331 gaps exceed sixty seconds, the longest being 48 hours. We resample onto a uniform ten-second grid and, critically, insert an explicit `is_gap` flag rather than interpolating across the long gaps. Interpolation would manufacture smooth synthetic behaviour over exactly the interval in which the machine was, in all likelihood, out of service;

a model trained on it would learn that the smoothest data is the most normal.

Analogue noise and sensor bias. Several analogue pressure channels report slightly negative values at rest, TP2 in 1,275,474 records. Where the resting value sits a few hundredths of a bar below zero, this is a calibration offset in the transducer zero point, and it is corrected by a per-channel offset estimated from idle-state records. Clipping at zero would be simpler, but it would destroy the noise structure needed to characterise the idle regime. The correction is guarded rather than applied blanket: a channel is offset-corrected only if its idle median already lies within 0.5 bar of zero. A below-zero minimum does not by itself establish a zero-point bias, because a channel that genuinely holds pressure when the unit is idle, as a cyclonic separator discharge line does, will have an idle median far from zero, and subtracting it would remove the physical signal along with the bias. The same statistic that identifies an artefact in one channel identifies real behaviour in another, and only domain reasoning separates them. A rolling median filter with a five-sample window suppresses transient spikes while preserving the sharp edges of state transitions, which a mean filter would smear.

Frozen channels. In 50,871 records (3.35%) every analogue channel is bit-identical to its predecessor. Sustained runs of this pattern indicate a held acquisition buffer. A machine that were truly holding steady would not produce runs of that length. They are flagged as suspect and excluded from training windows while retained for scoring, so the pipeline is never trained on artefact.

Digital debouncing. Digital flags are debounced with a minimum dwell time so that a single-sample flicker in COMP does not split one load cycle into three, which would corrupt every cycle-level feature downstream.

Management: the cycle store. Cleaned readings are aggregated into one record per compression cycle, retaining roughly 40 features: cycle duration, inter-cycle interval, duty ratio over trailing windows of 1, 6 and 24 hours, per-channel minimum, maximum, mean, standard deviation and slope within the cycle, pressure recovery rate, and digital flag counts. The result is written to columnar Parquet partitioned by month with dictionary encoding, which supports predicate pushdown on time ranges and cuts scan cost for the repeated train-test splits the evaluation requires. This is the largest single efficiency gain in the pipeline: 1.5 M rows become of order 10,000, a reduction of roughly two orders of magnitude, with no loss of the information that encodes a leak.

3.3 Modeling

Three tasks are specified because they consume different amounts of label information, and running all three quantifies how much the four scarce labels actually buy.

Task 1: Clustering for normal operating states. k -means and DBSCAN are applied to standardised cycle features; the wider taxonomy of anomaly types this task falls under is surveyed by Chandola et al. [4], and density-based

alternatives such as LOF [3] are held in reserve should DBSCAN’s global ϵ prove unsuitable across regimes. k -means is included for its interpretable centroids, which can be read as operating modes; DBSCAN is included because it does not assume convex, equally sized clusters and can leave points unassigned, which is the behaviour we want for cycles that fit no established mode. k is selected by silhouette score and the Davies-Bouldin index rather than by inertia alone. The expected result is a small number of regimes corresponding to idle, normal service load, and high demand; if a distinct cluster emerges that is populated disproportionately by pre-failure cycles, that is a finding in itself.

Task 2: Semi-supervised anomaly detection. Isolation Forest [9] and an LSTM autoencoder [11] are trained on February and March only, an interval containing no reported failure, and every later cycle is scored as deviation from that learned normality. This design is chosen specifically because the label set is incomplete: a fully supervised model would be taught that the unlabelled excursions in Figure 1 are normal, which is an assumption we have no evidence for. Isolation Forest is the primary method because it scales linearly, requires little tuning, and its path-length scores decompose by feature; the autoencoder is a secondary comparison, testing whether temporal dependencies beyond the engineered trailing windows carry additional signal. It is scheduled late deliberately, and it is the first item cut if earlier stages overrun, because no claim in this proposal depends on it.

Task 3: Lead-time classification. Each cycle is labelled by its hours-to-next-failure, and gradient-boosted trees are trained to predict whether a failure occurs within horizons of 6, 24 and 48 hours. Three horizons are used because the precursor analysis shows lead time varying by an order of magnitude across the four events, and a single horizon would conflate a model that cannot predict with a horizon that does not match the physics. Class weights address imbalance, probabilities are calibrated by isotonic regression so that a stated 0.8 means something operationally, and SHAP values [10] attribute each prediction to specific sensors.

3.4 Why not deep learning first

A tuned Isolation Forest over 40 interpretable cycle features that beats $F_1 = 0.32$ is a defensible, deployable result whose alarms can be explained to a maintenance planner. A deep model that beats it without an interpretable baseline for comparison establishes nothing about whether the gain came from representational capacity or from additional tuning effort. The deep model therefore appears as the comparison, and the interpretable baseline carries the headline.

4 EVALUATION PLAN

4.1 Effectiveness metrics

Accuracy is excluded deliberately. A model that predicts “normal” unconditionally achieves 98.02% accuracy on this dataset while being entirely useless, so accuracy cannot discriminate between methods here.

- **Precision, recall, F_1 and PR-AUC** at the cycle level, including for the duty-cycle baseline, which is re-evaluated at this grain so that the comparison is like for like. PR-AUC is preferred over ROC-AUC because under 2% positive prevalence, ROC curves are dominated by the true-negative mass and look optimistic [13].
- **Event-level detection rate.** An event counts as detected if any alarm fires within its window or within the lead-time horizon preceding it. With only four events this is a coarse measure, reported as a fraction with its exact binomial interval. A percentage would imply more precision than four samples support.
- **False alarms per 30 days** at a fixed alert budget. This is the number a maintenance planner will actually ask for, and it makes methods comparable at equal operational cost, not at equal threshold.
- **Lead time** in hours before onset, reported as a distribution across events, never as a bare mean.
- **If time permits: pattern support and confidence** for any association rules extracted from digital flag co-occurrence, with a minimum support threshold set so that a rule must hold across at least two distinct events to be reported. This guards against rules that describe a single incident. This is a secondary output and is not load-bearing for any claim in this proposal.

4.2 Efficiency metrics

- End-to-end wall-clock time to process all 1,516,948 rows, reported per stage, so that the bottleneck is identified before averaging hides it.
- Throughput in readings per second and peak resident memory, on fixed hardware with the specification stated.
- Per-cycle scoring latency, which must remain well below the mean inter-cycle interval of roughly eighteen minutes for the method to be deployable online.
- Model training time and serialised size, which bear on whether the method could run on constrained hardware at the vehicle instead of in a central system.

4.3 Experimental setup

Chronological splitting. The semi-supervised detector of Task 2 is fitted on February and March only, the longest interval containing no reported failure, and April through September is held out for scoring. The supervised lead-time model of Task 3 cannot use that split, because every positive episode falls in the held-out period; it is evaluated by leave-one-event-out instead, holding out a window around each event in turn. Random splitting is rejected outright for both. With ten-second sampling, adjacent records are near-duplicates, so a random split places near-identical rows on both sides of the partition and produces apparently excellent scores that measure nothing but leakage.

Blocked cross-validation. Model selection uses five-fold blocked cross-validation over contiguous time blocks, with a 24-hour purge window between folds so that no cycle's

trailing-window features overlap the boundary [2]. Standard k -fold is invalid here for the same reason random splitting is.

Baselines. Two must be beaten before any method is reported as successful: the naive duty-cycle threshold at $F_1 = 0.32$, and a seasonal-naive persistence model predicting that tomorrow resembles the same weekday last week. A method that does not clear both has not earned its complexity.

Ablations. Three, each answering a specific question: cycle-level against raw-sample features (does the aggregation help?); analogue-only against analogue-plus-digital (do the state flags contribute independently?); and trailing-window lengths of 1, 6 and 24 hours (what memory does the signal require?).

5 DISCUSSION AND TIMELINE

5.1 Anticipated challenges

Label incompleteness is the principal risk. Three quarters of high-duty-cycle hours carry no maintenance report. A model that detects those episodes will be penalised as imprecise even if it is correct, and there is no way to adjudicate from the data alone. No algorithmic fix exists for this, so the mitigation is procedural: unlabelled high-score episodes are reported as a separate category, with their sensor signatures characterised and compared against the signatures of confirmed events, instead of being silently counted as false positives. If they resemble confirmed leaks, that is evidence of under-reporting and is itself a contribution.

Four events is a very small sample. Any supervised result rests on four positive episodes, so variance estimates matter more than point estimates, and leave-one-event-out is the only defensible protocol.

Concept drift. Monthly mean duty cycle ranges from 0.059 to 0.297 across the record. A model trained on February normality may flag July normality as anomalous for reasons of seasonal demand. Mitigation is rolling re-estimation of the normal reference with a drift monitor on the score distribution.

Signal noise and calibration. Negative rest pressures and frozen acquisition buffers are handled as described above, but the residual risk is that a maintenance action, not a fault, produces the anomaly. The reports note maintenance following events 2, 3 and 4, and post-maintenance behaviour should be examined separately and never pooled into normal operation.

5.2 What we cannot judge in advance

Two of the choices above are guesses that the data will confirm or overturn, and we would rather name them now than defend them later.

The compression cycle is assumed to be a stable unit of analysis across both normal and faulty operation. If a fault changes the machine's mode of operation severely enough, that assumption fails and the cycle count inside the events could be too small to model. We have not verified this and will find out in week 1.

Table 2: Eight-week schedule, 21 June to 15 August 2027. Risk is front-loaded deliberately: if cycle segmentation is wrong, every downstream result is wrong, so it is validated in week 1.

Week	Dates (2027)	Activity and deliverable
1	21–27 Jun	Profiling, EDA, regime characterisation; cycle segmentation validated by hand
2	28 Jun–4 Jul	Cleaning, offset correction, filtering; Parquet cycle store versioned and locked
3	5–11 Jul	Clustering; k selected; operating-mode map delivered
4	12–18 Jul	Isolation Forest baseline on clean-window training; first event-level scores
5	19–25 Jul	Lead-time labelling, gradient boosting, blocked CV, SHAP attribution
6	26 Jul–1 Aug	Efficiency benchmarking on the full stream; LSTM autoencoder comparison if schedule allows
7	2–8 Aug	Ablations, sensitivity to alert budget, consolidation of results, drafting
8	9–15 Aug	Reserved buffer; final report, presentation and reproducible repository

The training window for the anomaly detector is assumed to be clean because no failure is reported in it. The label set is demonstrably incomplete, so the window may contain unreported faults; we will measure how many fall in it before relying on the result.

5.3 Backup plan

If cycle segmentation proves unreliable because the COMP flag is corrupted by the frozen-buffer artefact, the fallback is fixed 15-minute tumbling windows, which lose physical interpretability but require no state detection. If the cycle-level models fail to beat the baseline, the fallback is a change-point detector, specifically Bayesian online change-point detection, on the univariate hourly duty-cycle series. It is a weaker instrument, but it is robust, cheap to implement in two days, and guaranteed to produce a reportable result with a stated false-alarm rate.

5.4 Timeline

Table 2 allocates the eight weeks from 21 June to 15 August 2027, with week 8 held unassigned. The schedule reflects a judgement about where the uncertainty lies: weeks 1 and 2 are load-bearing, and a segmentation error discovered in week 6 would invalidate everything built on top of it. If weeks 5 or 6 overrun, the cut order is fixed in advance: association-rule mining first, then the LSTM comparison, then the ablation on trailing-window length. The cycle-level feature store, the clustering, the Isolation Forest baseline and the event-level evaluation are not negotiable, because the central claim of this proposal depends on them.

6 CONCLUSION AND FUTURE WORK

This proposal specifies a complete data mining pipeline over 1,516,948 sensor readings from an operational metro Air Production Unit, from profiling through to event-level evaluation. Its argument rests on three claims that profiling of

the raw data supports. First, the signal is strong and physically interpretable: compressor duty cycle rises 6.8-fold inside documented failure windows. Second, the problem is nonetheless hard for a reason that per-sample metrics conceal, namely that roughly three quarters of high-duty-cycle hours carry no failure report, so a naive detector reaches only $F_1 = 0.32$. Third, the compression cycle is the right unit of analysis, because it is the grain at which a leak physically manifests, and it reduces the working set by roughly two orders of magnitude at no informational cost.

The expected deliverables are cycle-level anomaly scores across all 213 days with event-level precision, recall and lead time reported against the four documented failures; a published comparison against the stated baseline; SHAP attributions naming the sensors behind each alarm; and a versioned, seeded repository that reruns end to end.

Several directions extend the work. A real-time dashboard consuming the streaming scorer would close the loop to operations, and the sub-eighteen-minute latency target is set with that in mind. Transfer to the sibling MetroPT-1 and MetroPT-2 releases [15], which cover different vehicles and include oil-leak failures, would test whether cycle-level features generalise across units and fault types. Finally, remaining useful life estimation is the natural successor: detection tells a planner that something is wrong, but a calibrated RUL estimate tells them how long they have, which is what schedules are actually built from.

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